

Šimon Hykl

Aircraft Design | Thermal Management

Location: Prague, Czech Republic
(committed to relocation)

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PROFESSIONAL SUMMARY

I am a recent graduate of aeronautical engineering at CTU Prague with a strong foundation in electrical engineering and additional academic experience at Delft University of Technology. Specialized in the design of classic small aircraft and compliance with European airworthiness standards (UL-2, CS-22, CS-23). 6 years of Formula Student experience with motorsport-focused mechanical design, thermal management, and CFD. Used to high-pressure teamwork in a fast-paced environment. Committed to international relocation.

EDUCATION

Czech Technical University in Prague, Faculty of Electrical Engineering Sep 2019 – Jun 2023

Bachelor's degree (B.Eng) in electrical engineering

Study program: Open Electronic Systems

Thesis: Scattering of an Electromagnetic Wave on a Two-Dimensionally Periodic Structure

Grade: B

Czech Technical University in Prague, Faculty of Mechanical Engineering Sep 2023 – Sep 2026

Master's degree (M.Eng) in aerospace engineering

Study program: Aircraft & Unmanned Systems

Thesis: Design of a Cooling System for a Formula Student Traction Accumulator

Grade: A

Delft University of Technology Sep 2025 – Jan 2026

ERASMUS+ (1 semester)

Aircraft avionics and operations, Advanced heat transfer, Non-linear dynamics, Design of Lightweight Structures, Multiphysics simulations

EXPERIENCE (FORMULA STUDENT)

eForce Prague Formula 2019 – 2025

Electrical Engineer, Mechanical Engineer

PCB hardware design and testing (initially), Design and integration of composite structures, Mechanical design of high-voltage batteries, Space-constrained mechanical layout, Battery thermal management, Cooling design, CFD, FEA, Data analysis

Successful development of Czechia's first self-driving Formula Student car
(2022–2023) Driver

(2023) Leader of the LV electronics department

(2025) Leader of the traction battery subgroup

(2025) Representation of the team at EXPO 2025 in Osaka, Japan

The team's world ranking: 7 (DV), 12 (EV)

TECHNICAL SKILLS

Engineering Analysis	: Internal and external CFD, Conjugate heat transfer, Experimental CFD validation, Thermo-electrical multiphysics, Data-driven cooling design, Design of experiments, Cost evaluation, Genetic algorithms, Neural networks
Simulation	: Ansys Fluent, Ansys Mechanical, Ansys SpaceClaim, COMSOL Multiphysics
CAD & Packaging	: Dassault 3DEXperience, CATIA V5, Autodesk Fusion 360, High-Density Mechanical Packaging
Electronics & Hardware	: Altium Designer, KiCAD, High-Voltage Traction Batteries, PCB Design and Testing
Programming & Tools	: MATLAB, C (basics), LaTeX, Git
Manual Skills	: Manufacturing of composites (wet lamination, prepregs, CFRP, GFRP, AFRP), Lathe and mill machining, 3D printing (UL 94 V-0 filaments), Laser welding, PCB assembly
Languages	: Czech (native), English (fluent – CEFR C2), German (intermediate), Japanese (elementary)